In [1]:
```python
import pandas as pd
```

In [3]:
```python
# TASK 1: IPL Match Results

ipl_data = pd.read_csv("IPL Matches 2008-2020.csv")
```

In [4]:
```python
print("TASK 1 - First 10 IPL Match Results")
print(ipl_data.head(10))
```

```
TASK 1 - First 10 IPL Match Results
       id        city        date player_of_match  \
0  335982   Bangalore  2008-04-18     BB McCullum
1  335983  Chandigarh  2008-04-19      MEK Hussey
2  335984       Delhi  2008-04-19     MF Maharoof
3  335985      Mumbai  2008-04-20      MV Boucher
4  335986     Kolkata  2008-04-20       DJ Hussey
5  335987      Jaipur  2008-04-21       SR Watson
6  335988   Hyderabad  2008-04-22        V Sehwag
7  335989     Chennai  2008-04-23       ML Hayden
8  335990   Hyderabad  2008-04-24       YK Pathan
9  335991  Chandigarh  2008-04-25   KC Sangakkara

                                        venue  neutral_venue  \
0                       M Chinnaswamy Stadium              0
1  Punjab Cricket Association Stadium, Mohali              0
2                            Feroz Shah Kotla              0
3                            Wankhede Stadium              0
4                                Eden Gardens              0
5                        Sawai Mansingh Stadium              0
6   Rajiv Gandhi International Stadium, Uppal              0
7            MA Chidambaram Stadium, Chepauk              0
8   Rajiv Gandhi International Stadium, Uppal              0
9  Punjab Cricket Association Stadium, Mohali              0

                         team1                        team2  \
0  Royal Challengers Bangalore        Kolkata Knight Riders
1              Kings XI Punjab          Chennai Super Kings
2             Delhi Daredevils             Rajasthan Royals
3               Mumbai Indians  Royal Challengers Bangalore
4        Kolkata Knight Riders              Deccan Chargers
5             Rajasthan Royals              Kings XI Punjab
6              Deccan Chargers             Delhi Daredevils
7          Chennai Super Kings               Mumbai Indians
8              Deccan Chargers             Rajasthan Royals
9              Kings XI Punjab               Mumbai Indians

                   toss_winner toss_decision                       winner  \
0  Royal Challengers Bangalore         field        Kolkata Knight Riders
1          Chennai Super Kings           bat          Chennai Super Kings
2             Rajasthan Royals           bat             Delhi Daredevils
```

```
3          Mumbai Indians         bat  Royal Challengers Bangalore
4          Deccan Chargers        bat        Kolkata Knight Riders
5          Kings XI Punjab        bat             Rajasthan Royals
6          Deccan Chargers        bat             Delhi Daredevils
7           Mumbai Indians      field          Chennai Super Kings
8         Rajasthan Royals      field             Rajasthan Royals
9           Mumbai Indians      field              Kings XI Punjab

    result  result_margin eliminator method    umpire1        umpire2
0     runs          140.0          N    NaN  Asad Rauf    RE Koertzen
1     runs           33.0          N    NaN  MR Benson     SL Shastri
2  wickets            9.0          N    NaN  Aleem Dar  GA Pratapkumar
3  wickets            5.0          N    NaN   SJ Davis      DJ Harper
4  wickets            5.0          N    NaN  BF Bowden    K Hariharan
5  wickets            6.0          N    NaN  Aleem Dar      RB Tiffin
6  wickets            9.0          N    NaN  IL Howell      AM Saheba
7     runs            6.0          N    NaN  DJ Harper  GA Pratapkumar
8  wickets            3.0          N    NaN  Asad Rauf      MR Benson
9     runs           66.0          N    NaN  Aleem Dar      AM Saheba
```

In [5]:
```python
# TASK 2: Mobile Expenses
mobile_expenses = pd.DataFrame({"Date": ["2026-09-01","2026-09-02","2026-09-03","2026-09-04","2026-09-05","2026-09-06","2026-0
"Expense": ["Mobile Recharge","Internet Data","Phone Case","Mobile Recharge","Screen Protector","Internet Data","Mobile Rechar
"Amount": [299,149,399,299,199,149,99]})
```

In [6]:
```python
mobile_expenses.to_excel("mobile_expenses.xlsx",index=False)
```

In [7]:
```python
expenses = pd.read_excel("mobile_expenses.xlsx")
```

In [8]:
```python
print("\nTASK 2 - Last 3 Mobile Expenses")
print(expenses.tail(3))
```

```
TASK 2 - Last 3 Mobile Expenses
         Date           Expense  Amount
4  2026-09-05  Screen Protector     199
5  2026-09-06     Internet Data     149
6  2026-09-07   Mobile Recharge      99
```

In [9]:
```python
# TASK 3: Zomato Restaurant Dataset
zomato_columns = ["Restaurant ID","Restaurant Name","Country Code","City","Address",
```

```
"Locality","Locality Verbose","Longitude","Latitude","Cuisines","Average Cost for two","Currency","Has Table booking",
"Has Online delivery","Is delivering now","Switch to order menu","Price range","Aggregate rating","Rating color",
"Rating text","Votes"]
```

In [10]:
```
zomato_data = pd.read_csv("Zomato.csv",sep="\t",header=None,names=zomato_columns,skiprows=1)
```

In [11]:
```
print("\nTASK 3 - Zomato Dataset Information")
zomato_data.info()
```

```
TASK 3 - Zomato Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 2339 entries, 0 to 2338
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Restaurant ID         2339 non-null   int64
 1   Restaurant Name       2339 non-null   str
 2   Country Code          2339 non-null   int64
 3   City                  2339 non-null   str
 4   Address               2339 non-null   str
 5   Locality              2339 non-null   str
 6   Locality Verbose      2339 non-null   str
 7   Longitude             2339 non-null   float64
 8   Latitude              2339 non-null   float64
 9   Cuisines              2323 non-null   str
 10  Average Cost for two  2339 non-null   int64
 11  Currency              2339 non-null   str
 12  Has Table booking     2339 non-null   int64
 13  Has Online delivery   2339 non-null   int64
 14  Is delivering now     2339 non-null   int64
 15  Switch to order menu  2339 non-null   int64
 16  Price range           2339 non-null   int64
 17  Aggregate rating      2339 non-null   float64
 18  Rating color          2339 non-null   str
 19  Rating text           2339 non-null   str
 20  Votes                 2339 non-null   int64
dtypes: float64(3), int64(9), str(9)
memory usage: 383.9 KB
```

In [12]:
```
print("\nTASK 3 - Statistical Description")
print(zomato_data.describe(include="all"))
```

```
TASK 3 - Statistical Description
        Restaurant ID Restaurant Name  Country Code       City  \
count    2.339000e+03            2339        2339.0       2339
unique            NaN            1483           NaN         30
top               NaN  Domino's Pizza           NaN  New Delhi
freq              NaN              63           NaN        436
mean     1.018529e+07             NaN           1.0        NaN
std      8.742341e+06             NaN           0.0        NaN
min      1.900000e+02             NaN           1.0        NaN
25%      9.687300e+04             NaN           1.0        NaN
50%      1.780580e+07             NaN           1.0        NaN
75%      1.859520e+07             NaN           1.0        NaN
max      1.893942e+07             NaN           1.0        NaN

                          Address Locality Locality Verbose    Longitude  \
count                        2339     2339             2339  2339.000000
unique                       1675      493              496          NaN
top     Sarwate Bus Stand, Indore    Sadar    Sadar, Nagpur          NaN
freq                           12       32               32          NaN
mean                          NaN      NaN              NaN    73.578877
std                           NaN      NaN              NaN    16.685124
min                           NaN      NaN              NaN     0.000000
25%                           NaN      NaN              NaN    73.807018
50%                           NaN      NaN              NaN    77.067312
75%                           NaN      NaN              NaN    78.404665
max                           NaN      NaN              NaN    88.489851

           Latitude      Cuisines  ...  Currency Has Table booking  \
count   2339.000000          2323  ...      2339       2339.000000
unique          NaN           908  ...         1               NaN
top             NaN  North Indian  ...       Rs.               NaN
freq            NaN           133  ...      2339               NaN
mean      19.850289           NaN  ...       NaN          0.036768
std        6.471913           NaN  ...       NaN          0.188232
min        0.000000           NaN  ...       NaN          0.000000
25%       17.493378           NaN  ...       NaN          0.000000
50%       19.000026           NaN  ...       NaN          0.000000
75%       22.699190           NaN  ...       NaN          0.000000
max       32.709480           NaN  ...       NaN          1.000000

        Has Online delivery  Is delivering now  Switch to order menu  \
```

```
count        2339.000000    2339.000000          2339.0
unique               NaN            NaN             NaN
top                  NaN            NaN             NaN
freq                 NaN            NaN             NaN
mean            0.604104       0.061137             0.0
std             0.489147       0.239633             0.0
min             0.000000       0.000000             0.0
25%             0.000000       0.000000             0.0
50%             1.000000       0.000000             0.0
75%             1.000000       0.000000             0.0
max             1.000000       1.000000             0.0

        Price range  Aggregate rating  Rating color Rating text         Votes
count   2339.000000       2339.000000          2339        2339   2339.000000
unique          NaN               NaN             7          10           NaN
top             NaN               NaN        5BA829   Very Good           NaN
freq            NaN               NaN           963         962           NaN
mean       2.171868          3.839461           NaN         NaN   1038.811885
std        0.833153          0.815160           NaN         NaN   1257.786474
min        1.000000          0.000000           NaN         NaN      0.000000
25%        2.000000          3.700000           NaN         NaN    183.500000
50%        2.000000          4.000000           NaN         NaN    697.000000
75%        3.000000          4.200000           NaN         NaN   1368.500000
max        4.000000          4.900000           NaN         NaN  11118.000000

[11 rows x 21 columns]
```

In [13]:

```
print("\nTASK 3 - Missing Values")
print(zomato_data.isnull().sum())
```

```
TASK 3 - Missing Values
Restaurant ID            0
Restaurant Name          0
Country Code             0
City                     0
Address                  0
Locality                 0
Locality Verbose         0
Longitude                0
Latitude                 0
Cuisines                16
Average Cost for two     0
Currency                 0
Has Table booking        0
Has Online delivery      0
Is delivering now        0
Switch to order menu     0
Price range              0
Aggregate rating         0
Rating color             0
Rating text              0
Votes                    0
dtype: int64
```

In [14]:
```
# TASK 4: Export First 20 Rows

filtered_data = ipl_data.head(20)
```

In [15]:
```
filtered_data.to_csv("filtered_data.csv",index=False)
```

In [16]:
```
print("\nTASK 4")
print("First 20 IPL records exported to filtered_data.csv")
```

```
TASK 4
First 20 IPL records exported to filtered_data.csv
```

In [18]:
```
# TASK 5: Export Dataset to Excel

ipl_data.to_excel("ipl_analysis.xlsx",sheet_name="Analysis2024",index=False)
```

```
print("\nTASK 5")
print("IPL dataset exported to ipl_analysis.xlsx")
print("Sheet name: Analysis2024")
```

```
TASK 5
IPL dataset exported to ipl_analysis.xlsx
Sheet name: Analysis2024
```

In [ ]: